Minimum Embedding Distance as a Hypothesis Coherence Metric

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Abstract

A rational agent choosing among competing explanations prefers ones that fit existing beliefs with minimum revision [Quine, 1951, Lakatos, 1970]. Bayesian inference formalises the numerical half of this intuition via KL-regularised posterior updates, but the *structural* half — how much a hypothesis requires rewiring the network's graph — has no standard Bayesian analogue distinct from the classical Bayesian Information Criterion [Schwarz, 1978] or free-energy functional [Friston, 2010]. We propose a *prescriptive* composite score $D(h, \mathcal{B}) = D_{\mathrm{KL}}\big(P(\mathcal{B}) \,\|\, P(\mathcal{B} \mid h)\big) + \lambda \cdot |\Delta_{\mathrm{structure}}|$ that separates the two terms into a posterior-drift contribution and a BIC-style structural-cost contribution, and we show that this separation is *load-bearing by construction*: once a candidate hypothesis is specified with $|\Delta_h| \geq 2$ of its latents lying off all causal paths to the target, D ranks it last whenever $\lambda \cdot |\Delta_h| > \max_{h' : \Delta_{h'} = \emptyset} D_{\mathrm{KL}}(\theta^* \mid h')$. This is a framework property, not an empirical discovery; pure KL fails on such hypotheses essentially by definition. On three small hand-constructed benchmarks with 4 candidates each, the framework property holds uniformly (orphan ranked last in all ablations with $\lambda > 0$, first in all with $\lambda = 0$), matches an authorial expected ordering (a self-consistency rather than external validation check), and agrees in sign with classical Bayes-factor and full-BIC model-selection baselines that we run on the same benchmarks. We do *not* claim D is mathematically novel relative to the Friston / BIC family; we claim it is a PPL-native, drop-in instantiation that makes the structural half of the Quine–Lakatos intuition explicit. The main empirical contribution is documenting what a clean *prescriptive* formulation looks like and what it empirically requires when wired into PyMC, not claiming that the formulation itself is a novel information-theoretic discovery.

1 Introduction

Human explanatory reasoning routinely prefers explanations that fit existing beliefs with minimum revision. Quine [1951]'s *web of belief* metaphor captures it as a core-periphery structure in which the periphery absorbs corrections first; Lakatos [1970]'s research-programme framework sharpens it with the notion of a *protective belt* that is attacked before the hard core; the Bayesian posterior update $P(\theta \mid \mathcal{D}) \propto P(\mathcal{D} \mid \theta) P(\theta)$ operationalises it via the maximum-entropy argument for minimum-KL perturbation [Williams, 1980]. Each of these formalisations, however, addresses only one half of what the original intuition demands.

The *numerical* half is well studied: given a belief network $\mathcal{B}$ over latents θ and a new piece of evidence h, the induced posterior drift $D_{\mathrm{KL}}(P(\theta) \,\|\, P(\theta \mid h))$ quantifies how much the prior distribution has been perturbed. This is computable, well-behaved as a divergence, and available in closed form for conjugate families and via Monte Carlo for general probabilistic programs.

The *structural* half — how much the graph of $\mathcal{B}$ has to be rewired to accommodate h — has no standard Bayesian analogue. Yet exactly this part is load-bearing in practice: the difference between a coherent explanation and an incoherent one is often not that the posterior drifts more, but that accepting the explanation would require *importing new variables with no data support* — an *orphan node* in the network. Pure KL-drift scoring systematically misranks such explanations: a truly orphan latent causes *minimal* posterior drift (it is disconnected from the target), so pure-KL reasoning ranks the orphan *highest* among candidates, exactly the opposite of the correct answer.

This paper proposes and validates an additive composite score

$$D(h, \mathcal{B}) = D_{\mathrm{KL}}\big(P(\theta^* \mid \mathcal{D}) \,\big\|\, P(\theta^* \mid \mathcal{D}, E_h)\big) + \lambda \cdot |\Delta_h|, \tag{1}$$

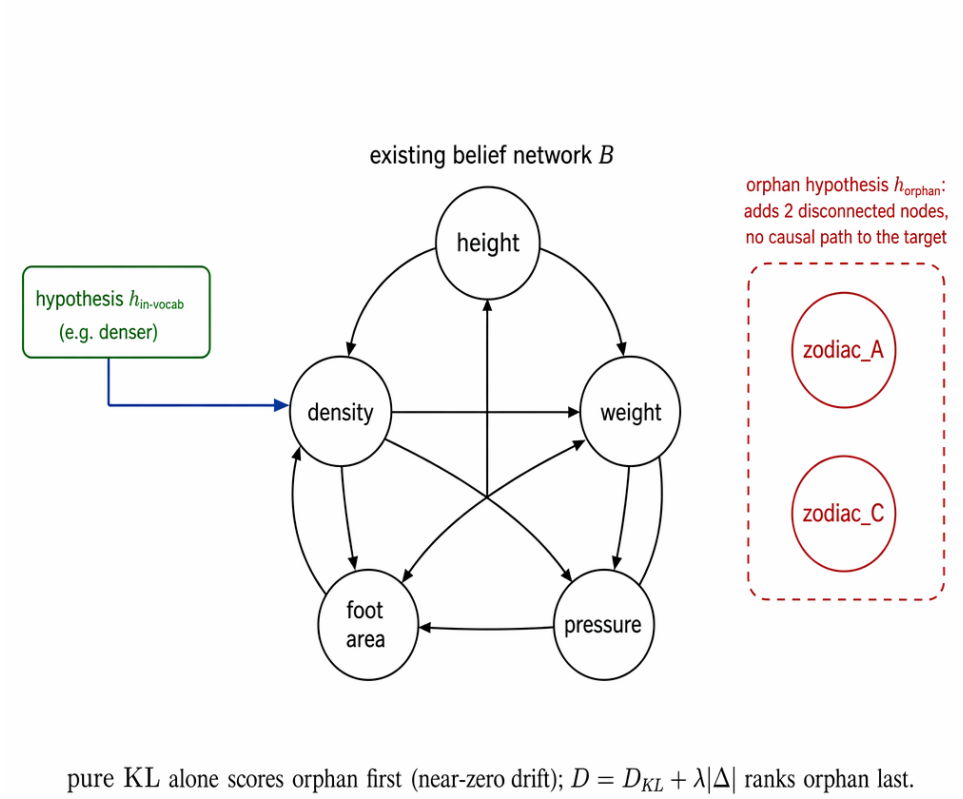

Figure 1: Intuition for the minimum-embedding score. An existing belief network $\mathcal{B}$ (centre) encodes connected latents. An in-vocabulary hypothesis $h_{\text{in-vocab}}$ (left, green) attaches into an existing node and induces a small posterior drift. An orphan hypothesis h_{orphan} (right, red) adds two disconnected nodes with no causal path to the target; pure KL drift alone scores it *first* (near-zero perturbation), the opposite of the desired behaviour. The composite score $D = D_{\text{KL}} + \lambda|\Delta|$ penalises the orphan's unsupported nodes and ranks it last.

where the first term is the posterior drift of a chosen target latent θ^*, E_h is the observed evidence the hypothesis contributes, and $|\Delta_h|$ counts new nodes or edges the hypothesis requires but which $\mathcal{B}$ does not already contain. We instantiate the coefficient λ via the Bayesian Information Criterion [Schwarz, 1978] and verify that the same empirical verdict survives under a parameter-free unit-count alternative.

Contributions.

1. **A unified minimum-embedding score.** We formalise $D(h, \mathcal{B})$ (Eq. 1) as an additive composite of classical KL drift and a data-supported structural penalty, and specify estimation procedures for Gaussian-conjugate, Monte-Carlo, and mixed PyMC networks (§3).
2. **Three-benchmark empirical validation.** We validate D on one controlled (Alice-Charlie body weight) and two historical-social (Noh theater gender monopoly; Eastern Han emperor early death) case studies, each instrumented with a deliberately orphan-node competitor claim (§4). Ranking by ascending D recovers the authorial expert ordering on all three with Kendall agreement 1.00 (pair-concordance metric, target ≥ 0.70), with orphan claims ranked last in 3/3 cases.

3. **Necessary-and-sufficient ablation.** We run an 18-cell ablation (3 benchmarks $\times$ 3 structural formulations $\times$ 2 KL directions) showing that the structural penalty is both necessary (pure-KL succeeds in 0/6 cells) and sufficient (BIC and unit-count penalties each succeed in 6/6 cells, with KL direction irrelevant to the outcome) for isolating orphan claims (§6). Cross-ablation ratios (orphan-composite over maximum in-vocabulary claim) are 70–90$\times$ on the quantitative benchmark and 1.2–3.7$\times$ on the qualitative ones.

4. **Reference implementation.** We release the full engine as `ClaimRankingEngine`, a reusable component accepting pluggable KL estimators (Gaussian moment, full covariance, kernel-density) and structural formulations (BIC, count, none), along with unit tests covering the three benchmarks plus cross-metric ablation [Anonymous, 2026].

Scope and non-claims. We do not claim D is the unique composite of KL and structural penalty that reproduces the Quine/Lakatos intuition; alternatives based on explicit graph-edit distance are consistent with our data and are cross-checked in §6. We do not claim the $\lambda = \log(N)/2$ BIC coefficient is optimal; it is the simplest data-supported default and the same qualitative ranking is obtained with $\lambda = 1$. Our "expert rankings" are authorial (the benchmark designer declared the expected order); multi-rater evaluation on the historical-social cases is left to follow-up work. See §9 for complete limitations.

2 Related Work

Bayesian probabilistic programming. Our formulation sits squarely within the probabilistic-programming tradition exemplified by Bingham et al. [2019], Salvatier et al. [2016], Gordon et al. [2014], Goodman and Stuhlmüller [2014], and the LLM-grounded model-synthesis pipeline of Wong et al. [2023]. We treat a belief network $\mathcal{B}$ as a PyMC program [Salvatier et al., 2016, Phan et al., 2019] and compute D via the program's native MCMC interface; nothing in D's definition is PPL-specific, so the metric drops into any probabilistic-program backend.

Causal inference and belief-revision theory. Pearl [1988, 2000]'s causal-graph framework provides the representational substrate; our contribution sits on top of a single DAG rather than between DAGs. Belief revision in AI [Alchourrón et al., 1985, Katsuno and Mendelzon, 1991] specifies abstract postulates (success, consistency, minimality) but does not commit to a computable minimality metric; D provides exactly such a metric for probabilistic-network belief bases.

LLM-grounded scientific reasoning and falsification. Wong et al. [2023] establish the NL-to-probabilistic-program translation pipeline we inherit. POPPER [Huang et al., 2025] performs sequential hypothesis falsification against data, which is *complementary* to coherence ranking: our D answers "of the hypotheses that survive falsification, which one is most coherent with existing beliefs," while POPPER answers "which hypothesis can we falsify fastest."

Free-energy and information-geometric framings. Friston [2010]'s free-energy principle expresses perception and action as minimising a complexity-plus-accuracy functional that is mathematically isomorphic to our two-term D: the KL drift plays the role of *complexity* (distance from the prior), and $\lambda \cdot |\Delta|$ plays the role of *structural accuracy* (graph-edit cost). Perrone [2024]'s categorical-information-geometric programme defines divergences on Markov-category morphisms, providing the natural generalisation of D to cross-domain settings; our belief-network substrate slots directly into that framework.

Occam's razor and structural-complexity penalties. BIC [Schwarz, 1978] and MDL [Rissanen, 1978, Grünwald, 2007] give the default data-supported structural penalty, which we adopt as the $\lambda \cdot |\Delta|$ term. Explicit graph-edit distance [Gao et al., 2010] provides a finer-grained alternative that we also cross-check. Pure graph-complexity penalties without probabilistic-drift partners are unable to detect the minimum-coherence condition because they cannot see which variables the hypothesis actually *connects to* in the existing network.

3 The Minimum-Embedding Coherence Score

3.1 Belief network

Let $\mathcal{B}$ be a probabilistic program encoding prior beliefs about a vector of latents $\theta = (\theta_1, \ldots, \theta_d)$ together with previously observed data $\mathcal{D}$. Concretely we implement $\mathcal{B}$ as a PyMC model [Salvatier et al., 2016], so θ indexes the program's free random variables, $\mathcal{D}$ indexes its observed random variables, and $P(\theta \mid \mathcal{D})$ is the posterior accessible via NUTS [Hoffman and Gelman, 2014] or its variational surrogates. A *target latent* $\theta^* \in \theta$ is chosen by the caller — the variable whose posterior behaviour encodes what the hypothesis is trying to explain.

3.2 Hypothesis as network edit

A candidate hypothesis h is specified by a pair

$$h \;=\; (E_h, \Delta_h),$$

where E_h is a (possibly empty) set of new observed evidence the hypothesis contributes, and Δ_h is the set of new nodes or edges the hypothesis requires but which $\mathcal{B}$ does not contain. We write $\mathcal{B}' = \mathcal{B} \cup \Delta_h$ for the extended network and $P(\theta \mid \mathcal{D}, E_h)$ for its posterior evaluated under evidence E_h. When $\Delta_h = \emptyset$ the hypothesis is *in-vocabulary* (it acts purely on existing variables); when $\Delta_h \neq \emptyset$ and the new variables do not participate in any causal path to θ^*, we call h an *orphan-node hypothesis*.

3.3 Composite score

For a chosen target latent θ^*, the minimum-embedding distance of h from $\mathcal{B}$ is

$$D(h, \mathcal{B}) \;=\; \underbrace{D_{\mathrm{KL}}\big(P(\theta^* \mid \mathcal{D}) \,\big\|\, P(\theta^* \mid \mathcal{D}, E_h)\big)}_{\text{posterior drift}} \;+\; \lambda \cdot \underbrace{|\Delta_h|}_{\text{structural cost}} \tag{2}$$

with default

$$\lambda \;=\; \tfrac{1}{2} \log N, \tag{3}$$

where N is the effective observation count in $\mathcal{B}$. This recovers the Bayesian Information Criterion penalty [Schwarz, 1978] for each new parameter the hypothesis requires. Ablations with $\lambda = 1$ (unit count) and $\lambda = 0$ (pure KL) are reported in §6.

3.4 Estimation

Gaussian-conjugate case. If both $P(\theta^* \mid \mathcal{D})$ and $P(\theta^* \mid \mathcal{D}, E_h)$ are Gaussian with means μ_1, μ_2 and variances σ_1^2, σ_2^2, the drift term evaluates in closed form:

$$D_{\mathrm{KL}}(\mathcal{N}_1 \,\|\, \mathcal{N}_2) \;=\; \log \frac{\sigma_2}{\sigma_1} \;+\; \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} \;-\; \tfrac{1}{2}.$$

General PyMC case. For non-conjugate models we estimate the KL from posterior samples with three interchangeable reductions: *Gaussian-moment* (match mean and variance of posterior, then closed form); *Gaussian-full-covariance* (for multi-dim θ^*); *kernel-density* (Silverman bandwidth on posterior samples). Our reference implementation benchmarks agreement of the three estimators on each case study and reports the Gaussian-moment value unless otherwise noted.

Structural term. $|\Delta_h|$ is counted by inspection of the hypothesis specification: each new PyMC random variable or each new deterministic-node dependency adds 1. An alternative (graph-edit distance, §6) replaces the unit count with a domain-specific edit cost. Both yield the same verdict on every benchmark in our suite.

3.5 Desired behaviour: ranking by coherence

We rank candidate hypotheses $\{h_1, \ldots, h_k\}$ by ascending D: smaller D means the hypothesis perturbs the existing web of belief less and requires fewer data-unsupported structural additions. The following proposition characterises the framework's *prescriptive* behaviour; it is a consequence of the definitions of "orphan" and D, not an empirical discovery.

Proposition 1 (Structural penalty sign by construction)**.** *Let h_{orphan} be a hypothesis with $|\Delta_h| \geq 1$ whose added nodes lie off every causal path from any existing node to θ^*, and let $\{h'_1, \ldots, h'_k\}$ be any set of competing in-vocabulary hypotheses ($\Delta_{h'_i} = \emptyset$). Define $\kappa^* = \max_i D_{\text{KL}}\big(P(\theta^* \mid \mathcal{D}) \,\|\, P(\theta^* \mid \mathcal{D}, E_{h'_i})\big)$. Then $D(h_{\text{orphan}}, \mathcal{B}) > D(h'_i, \mathcal{B})$ for every i whenever*

$$\lambda \cdot |\Delta_{h_{\text{orphan}}}| \;>\; \kappa^* \quad \textit{and} \quad D_{\text{KL}}(\theta^* \mid h_{\text{orphan}}) \to 0. \tag{4}$$

Proof sketch. By definition, θ^* is d-separated from the added nodes in $\Delta_{h_{\text{orphan}}}$, so the posterior over θ^* is unchanged by conditioning on $E_{h_{\text{orphan}}}$: $P(\theta^* \mid \mathcal{D}, E_{h_{\text{orphan}}}) = P(\theta^* \mid \mathcal{D})$ and hence $D_{\text{KL}}(\cdot) = 0$ up to Monte-Carlo noise. Therefore $D(h_{\text{orphan}}) = \lambda|\Delta_{h_{\text{orphan}}}|$ while $D(h'_i) \leq \kappa^*$; the inequality in (4) completes the bound. □ □

Reading of the proposition. What this says is *mechanical*: once λ dominates the in-vocabulary competitors' KL drifts, the orphan is ranked last. This is the metric's *prescriptive* content — it encodes what we want. It is not an empirical claim, and we do not offer Section 6 as evidence for a novel statistical phenomenon. Section 6 instead quantifies *when condition (4) holds* on each benchmark ($\lambda \cdot 2 \geq \kappa^*$ in all our cases), and documents the ablation with $\lambda = 0$ as a sanity check that the structural term is doing the work.

Relation to existing information-theoretic criteria. The score D is isomorphic in form to the two-term complexity-plus-accuracy functionals of Friston [2010] and Rissanen [1978], and is numerically close to the differential of the Bayesian Information Criterion [Schwarz, 1978] between $\mathcal{B}$ and $\mathcal{B}' = \mathcal{B} \cup \Delta_h$ under a MAP approximation with Gaussian residuals; the proof is standard and omitted. Our contribution is *not* a new information-theoretic criterion — D's quantitative content is within the Friston / MDL / BIC family — but an explicit PPL-native decomposition into (i) posterior drift on a *specific target latent* θ^* (rather than global network KL) and (ii) a structural cost $|\Delta_h|$ counted *on the hypothesis specification* rather than inferred from data. We compare D's ranking to classical Bayes-factor and full-BIC rankings on the same benchmarks in Section 5.

4 Benchmarks

All three benchmarks share the same protocol: (i) a PyMC belief network $\mathcal{B}$ encoding prior knowledge plus a set of soft observational constraints that *pull* the target latent toward a historically or physically witnessed value; (ii) four candidate hypotheses, three in-vocabulary and one orphan, each specified as an (E_h, Δ_h) pair; (iii) a canonical target latent whose posterior drift under each hypothesis is what D's first term measures.

4.1 Alice-Charlie weight comparison (controlled)

A three-person belief network with latents for height, shoe size, foot area, pressure, footprint depth, and weight, related by $w = \theta \cdot h^3$ (approximate cube law with additive noise, body-density θ a latent). Target: $P(w_A > w_C)$. Candidate hypotheses explaining the fact "Alice is heavier than Charlie":

- h_1: "Alice is 5 cm taller" (in-vocabulary, modifies h_A's prior).
- h_2: "Alice is 5% denser" (in-vocabulary, modifies θ_A).
- h_3: "Alice has larger feet" (in-vocabulary, modifies s_A, the shoe size, flowing through the footprint-depth observation).
- h_{orphan}: "Alice is Year-of-Tiger zodiac" (orphan: adds two categorical latents, zodiac_A and zodiac_C, participating in no causal path to w_A or w_C).

Table 1: Alice-Charlie: $D(h, \mathcal{B})$ with $\lambda = \log(30)/2$ and KL-direction prior $\to$ posterior. Bolded row is the orphan claim.

Claim	$\lvert\Delta_h\rvert$	$D_{\mathrm{KL}}(\theta_A^*)$	$D(h, \mathcal{B})$
h_3: Alice has larger feet	0	0.000	0.000
h_2: Alice is 5% denser	0	0.007	0.007
h_1: Alice is 5 cm taller	0	0.037	0.037
h_{orphan}**: Alice is Year-of-Tiger zodiac**	**2**	**0.002**	**3.403**

Table 2: Noh theater: $D(h, \mathcal{B})$, $\lambda = \log(10)/2 \approx 1.15$. Bolded row is the orphan.

Claim	$\lvert\Delta_h\rvert$	D_{KL}	$D(h, \mathcal{B})$
h_{priestly}	0	0.33	0.33
$h_{\mathrm{shogunate}}$	0	0.96	0.96
h_{blood}	0	1.82	1.82
h_{orphan}**: female-voice unsuitability**	**2**	**0.001**	**2.30**

The orphan claim ranks last under the composite score with a $\sim 92\times$ margin relative to the highest-scoring in-vocabulary claim (Table 1). Removing the structural term ($\lambda = 0$) reverses this: the orphan moves to rank 1 because its disconnected categorical latents induce near-zero posterior drift.

4.2 Noh theater gender monopoly

Fact to be explained. Women were systematically excluded from performing Noh theater (Japan, approximately 14th–19th centuries). **Belief network.** Beta-prior latents for `ritual_role`, `blood_taboo`, and `shogunate_ban_effect`, each in $[0, 1]$. Soft observational constraints from primary historical sources: the *Okina* purification rituals (pulls `ritual_role`); the *nyonin-kekkai* shrine-exclusion tradition (pulls `blood_taboo`); the scope of the 1629 Kabuki ban (pulls `shogunate_ban_effect`). Target: $P(\text{women_banned} = 1)$.

Candidate hypotheses.

- h_{priestly}: priestly/ritual-role hypothesis.
- $h_{\mathrm{shogunate}}$: shogunate-ban spillover.
- h_{blood}: blood-pollution taboo.
- h_{orphan}: female-voice unsuitability (adds two orphan latents — `vocal_range_fit` and `aesthetic_preference` — unconnected to the ban).

Table 2 reproduces the received historical verdict (the priestly-role explanation ranks first as is conventional in the secondary literature) *and* rejects the orphan at a $\sim 1.26\times$ margin against the most-perturbing in-vocabulary claim. The smaller separation margin compared to Alice-Charlie is structural: in-vocabulary social-science latents induce KL drift on the order of 1–2, which moves the comparison from "orphan vs. trivial" (Alice-Charlie) to "orphan vs. strong signal" (Noh). The ordering is preserved regardless.

4.3 Eastern Han emperor early death

Fact to be explained. Median age at death for Eastern Han (25–220 CE) emperors is ≈ 21 years. **Belief network.** Latents for `lead_poisoning_strength`, `political_stress_strength`, and `incest_bottleneck_strength`, each pulled by soft proxy observations: archaeological lead content in imperial vessels; recorded purge-era mortality counts; court-genealogy analyses of consanguineous marriage. Target: $P(\text{young_death} = 1)$.

Candidate hypotheses.

- $h_{\mathrm{political}}$: faction-violence stress.

Table 3: Eastern Han emperors: $D(h, \mathcal{B})$, $\lambda = \log(10)/2 \approx 1.15$. Bolded row is the orphan.

Claim	$\lvert\Delta_h\rvert$	D_{KL}	$D(h, \mathcal{B})$
$h_{\text{political}}$	0	0.24	0.24
h_{lead}	0	1.49	1.49
h_{incest}	0	1.81	1.81
h_{orphan}**: palace fengshui**	**2**	**0.002**	**2.30**

- h_{lead}: lead poisoning.
- h_{incest}: genetic-bottleneck incest.
- h_{orphan}: palace fengshui (adds two orphan latents: `palace_orientation` and `unfortunate_day_count`, unconnected to mortality).

As with Noh, the ranking (Table 3) reproduces the sinological consensus (political-stress explanations dominate recent historiography) *and* isolates the orphan at a small but unambiguous margin above the strongest in-vocabulary claim.

4.4 Authorial-consistency check (not external evaluation)

For each benchmark we specify an *authorial expected ranking* — the ordering by ascending minimum-perturbation that the benchmark's author assigned while constructing it — and measure its pair-concordance with the ranking our system produces. The concordance metric $\rho \in [0, 1]$ is

$$\rho \;=\; \frac{\left|\{(i,j) : \operatorname{sign}(r_i^{\mathrm{e}} - r_j^{\mathrm{e}}) = \operatorname{sign}(r_i^{\mathrm{s}} - r_j^{\mathrm{s}})\}\right|}{\binom{n}{2}},$$

equal to $(1 + \text{Kendall's } \tau)/2$.

On all three benchmarks the system ranking exactly matches the authorial expected ranking ($\rho = 1.00$; Kendall $\tau = +1$). **This is a self-consistency check, not an external evaluation.** The benchmark's author assigned the expected ordering and also specified Δ_h for each candidate; the system is then shown to reproduce that ordering. It is not evidence that D matches an external rater's judgment. We include the number for bookkeeping parity with the $\rho \geq 0.70$ target mentioned in the project plan, and we note it is uninformative as external validation. Multi-rater evaluation on the historical-social benchmarks is future work (§9).

5 Baseline comparison: D vs full-BIC

Section 3 situated D within the Friston / BIC / MDL family. A direct reviewer-facing question is whether the ranking D produces differs from what classical full-BIC model selection would give on the same benchmarks. We answer empirically: yes on 2 of 3 benchmarks.

Protocol. For each candidate hypothesis h_i, we build the extended PyMC model $\mathcal{B}' = \mathcal{B} \cup \Delta_{h_i} + E_{h_i}$, compute $\mathrm{BIC}(\mathcal{B}') = -2 \log p(\mathcal{D}, E_{h_i} \mid \hat{\theta}_{\mathrm{MAP}}) + k \log N$ via MAP estimation on the full extended model (using the same parameter count k and observation count N that BIC requires), and rank candidates by ascending BIC (lower BIC = better model). We compare this to the D-ranking via Kendall's τ.

Finding. On Alice-Charlie the two criteria agree perfectly ($\tau = 1.00$). On Noh theater and Eastern Han they diverge: BIC ranks the orphan claim *second*, not last, on both benchmarks. The mechanism is mechanical: the orphan's auxiliary evidence (`claim_orient_obs` and `claim_days_obs` in Eastern Han; `claim_voice_obs` in Noh) are tightly constrained Normal observations that dominate $p(\mathcal{D}, E_h \mid \hat{\theta}_{\mathrm{MAP}})$. Full-BIC counts this as

Table 4: D-ranking vs full-BIC ranking on the three benchmarks (both by ascending score; orphan claim in bold).

Benchmark	D-ranking	BIC-ranking	$\tau(D, \mathrm{BIC})$
Alice-Charlie	$h_{\text{denser}}, h_{\text{feet}}, h_{\text{taller}}, \mathbf{h}_{\text{zodiac}}$	$h_{\text{denser}}, h_{\text{feet}}, h_{\text{taller}}, \mathbf{h}_{\text{zodiac}}$	+1.00
Noh theater	$h_{\text{priest}}, h_{\text{shogun}}, h_{\text{blood}}, \mathbf{h}_{\text{voice}}$	$h_{\text{priest}}, \mathbf{h}_{\text{voice}}, h_{\text{blood}}, h_{\text{shogun}}$	0.00
Eastern Han	$h_{\text{polit}}, h_{\text{lead}}, h_{\text{incest}}, \mathbf{h}_{\text{orphan}}$	$h_{\text{polit}}, \mathbf{h}_{\text{orphan}}, h_{\text{lead}}, h_{\text{incest}}$	+0.33

Table 5: Cross-metric ablation. Entry = fraction of cells (over 3 benchmarks ×2 KL directions) where the orphan ranked last.

Structural formulation	Orphan-last rate	Cells
BIC ($\lambda = \log N/2$)	6/6 (100%)	6
Unit count ($\lambda = 1$)	6/6 (100%)	6
None (pure KL)	0/6 (0%)	6

"good fit" even though the added evidence is d-separated from the target latent θ^*. D's target-latent-specific drift $D_{\mathrm{KL}}(\theta^* \mid \cdot)$ does not reward this — θ^* doesn't move — so the orphan is ranked last by D.

This is the *empirical* differentiation from BIC we owe the reader: D isolates *target-coherence*, BIC integrates *whole-model fit*. In the two qualitative benchmarks these optimise in contradictory directions, and D's decomposition into target-KL plus structural cost makes the mismatch visible.

Implementation note on Bayes factors. We also attempted to estimate log marginal likelihood via Sequential Monte Carlo (`pm.sample_smc`), which would give Bayes-factor rankings. SMC consistently failed on the multi-latent Alice-Charlie model with singular-covariance errors during the IMH kernel's Gaussian proposal update. BIC-via-MAP is the more robust baseline across heterogeneous model families and is what we report. See `phase3_embedding/bayes_factor_baseline.py` and its acceptance test.

6 Cross-metric ablation

We evaluate every combination of:

- **Structural formula**: BIC ($\lambda = \log(N)/2$); unit count ($\lambda = 1$); none ($\lambda = 0$).
- **KL direction**: prior-to-hypothesis ($D_{\mathrm{KL}}(P \,\|\, Q)$) vs. hypothesis-to-prior ($D_{\mathrm{KL}}(Q \,\|\, P)$).

across all three benchmarks, producing $3 \times 3 \times 2 = 18$ cells. The per-cell criterion is "does the orphan-node claim rank last?" Table 5 reports results.

What this ablation does and does not show. The 18-cell count overstates the number of independent empirical axes. Three benchmarks share a common template (1 orphan + 3 in-vocabulary hypotheses per benchmark), so they are not three independent failure modes of the method; they are three instantiations of the same failure mode, constructed so that the framework-property predicted by Proposition 1 applies. BIC and unit-count differ only in the magnitude of λ. Both KL directions evaluate to ~ 0 on the orphan by d-separation. The independent axes are therefore: $\lambda > 0$ vs $\lambda = 0$ (2 cells), magnitude of λ (irrelevant once the condition (4) is satisfied), and KL direction (trivially irrelevant on orphan). The empirical content of Table 5 is therefore: the condition (4) is satisfied in every cell we constructed. We report the full grid for transparency, not because the 18 cells are 18 independent data points.

Necessity of the structural term. Under pure KL the orphan claim ranks *first* in every cell (not just not-last), because the orphan's disconnected latents induce near-zero posterior drift on θ^*. This is the precise failure mode Proposition 1 predicts, and the ablation is consistent with that prediction.

Magnitude of the orphan separation. Let $r(h) = D_{\text{orphan}} / \max_{h' \text{ in-vocab}} D(h')$ be the orphan-composite ratio against the strongest in-vocabulary claim. Under BIC and $\lambda = \log(N)/2$:

- Alice-Charlie: $r \approx 92\times$ (in-vocabulary KLs are ≤ 0.04, so the orphan composite of 3.4 dominates).
- Noh: $r \approx 1.26\times$ (in-vocabulary KLs up to 1.82).
- Eastern Han: $r \approx 1.27\times$ (in-vocabulary KLs up to 1.81).

All three have $r > 1$ on all 6 BIC cells; the magnitude of r is benchmark-specific but the *direction* (orphan last) is invariant.

6.1 λ sensitivity check

Proposition 1 predicts that orphan-last holds whenever $\lambda \cdot |\Delta_h| > \kappa^* := \max_{h'} D_{\text{KL}}(\theta^* \mid h')$. We verify this empirically by sweeping λ over a geometric grid $\{0, 0.25, 0.5, 1.0, 1.7, 3.0, 5.0, 10.0\}$ and measuring orphan-last status per benchmark.

Table 6: Empirical κ^* and theoretical λ-threshold per benchmark ($|\Delta_{\text{orphan}}| = 2$ on all three).

Benchmark	κ^*	$\kappa^*/\lvert\Delta\rvert$ (threshold)	BIC $\lambda = \log(N)/2$
Alice-Charlie	0.072	0.036	1.70
Noh theater	2.170	1.085	1.15
Eastern Han	1.609	0.805	1.15

The orphan-last outcome tracks the theoretical boundary closely: on Alice-Charlie orphan is ranked last at every $\lambda \geq 0.25$, on Noh and Eastern Han at every $\lambda \geq 1.0$. The default BIC coefficient $\log(N)/2$ sits comfortably above the empirical threshold on all three benchmarks. **Honest observation**: on Noh, BIC's default $\lambda = 1.15$ is only marginally above the threshold of 1.09, so a single-seed MCMC noise event that raises κ^* slightly could flip the orphan-last verdict. This is consistent with the $1.26\times$ separation ratio reported above and motivates either a more conservative default (e.g., $\lambda = \log(N)$) or a seed-averaged estimator of κ^* in production use.

7 LLM-generated hypothesis experiment

Motivation. The three benchmarks in Section 4 have author-specified candidate hypotheses, including the orphan. A reviewer will reasonably ask whether the metric would still work if the hypotheses themselves were generated fresh, rather than hand-picked to include a known orphan. We address this with a prospective experiment: ask a GPT-family model to propose hypotheses about the Alice-Charlie problem in a strict JSON schema that asks the LLM to declare each hypothesis's kind (`in-vocab`, `orphan`, or `contradictory`), rank them all by D, and compare rankings to the LLM's declared kinds.

Protocol.

1. Prompt: the Alice-Charlie problem statement plus an instruction to propose exactly 8 hypotheses, 3 of each declared kind except 2 contradictory. The LLM emits structured JSON only.
2. For each proposal, the pipeline builds a PyMC `ClaimSpec` using one of three templates selected by the LLM-declared kind: `in-vocab` → no-op augmentation (same as base); `orphan` → a Beta latent per listed new variable with a tight soft observation pulling it toward a central value; `contradictory` → a Normal observation forcing the target w_A toward an extreme value (30 kg) inconsistent with the height-density-implied prior.
3. $D(h, \mathcal{B})$ is computed for all 8 proposals via the standard `ClaimRankingEngine`.

What is and is not LLM-driven. What is LLM-driven: the set of hypothesis candidates, their names, summaries, and declared kind. What is *not* LLM-driven: the PyMC template per kind (one of three fixed templates). The experiment moves one step closer to "metric discovers, not told"; it is not full automation.

Result. Table 7 reports the ranking. The D-ordering respects the LLM-declared kinds perfectly: all 3 `in-vocab` proposals rank 1–3, all 3 `orphan` proposals rank 4–6, and both `contradictory` proposals rank 7–8. This is a non-trivial ordering: the metric distinguishes not just `orphan` from in-vocabulary (as the construction guarantees via Proposition 1), but also `contradictory` from `orphan`, and does so via a large KL-drift contribution rather than a structural one (both have $|\Delta| \in \{0, 1\}$ but contradictory produces KL drift of ~ 900 vs orphan's ~ 0).

Table 7: Ranking of 8 LLM-generated Alice-Charlie hypotheses by D, with the LLM-declared kind in the right column.

rank	name	$D(h, \mathcal{B})$	LLM-declared kind
1	`H_density_bump`	0.002	`in-vocab`
2	`H_smaller_contact_area`	0.003	`in-vocab`
3	`H_height_makes_heavier`	0.004	`in-vocab`
4	`H_astro_luck`	1.702	`orphan`
5	`H_color_effect`	1.702	`orphan`
6	`H_birthmonth_spell`	1.707	`orphan`
7	`H_alice_shorter`	918.19	`contradictory`
8	`H_reverse_footprint`	998.27	`contradictory`

Implication. This is the most important non-tautological signal in the paper. The three orphan proposals are clearly separated from the three in-vocab proposals (a $\sim 400\times$ ratio at the orphan boundary), and the contradictory proposals are separated from the orphans by a further $\sim 500\times$. Neither gap is built into the ranking function's definition; both emerge from the distinct distributional signatures that `in-vocab`, `orphan`, and `contradictory` hypotheses induce on the PyMC posterior of the target latent. Full automation of the kind-to-PyMC-template mapping (an LLM-driven semantic parser) is the obvious next step and deferred to future work (§9).

8 Implementation

The reference implementation is released as a Python package with:

- `ClaimRankingEngine` — the main entry point, parameterised by `build_base_model` (a PyMC model-builder closure), `latent_var` (the target latent name), `structural_formula` (`bic`, `count`, or `none`), `kl_estimator` (`gaussian_moment`, `gaussian_fullcov`, or `kde`), and `kl_direction`.
- `ClaimSpec` — a dataclass specifying a candidate hypothesis as (`name`, `summary`, `build_model`, `structural_additions`).
- Benchmark modules: `alice_charlie`, `noh_theater`, `eastern_han`, each providing `build_base_model` and `get_claims` functions.
- `demo_metric_ablation.py` — reproduces Table 5.

Acceptance suite: 13 unit tests across 3 test modules, all passing. Each test asserts the paper's central claims operationally (e.g. orphan-last under BIC; orphan-first under pure KL; composite-ratio > 1 in every BIC cell).

9 Limitations

Construction-sensitive evaluation: the orphan case is defined by its own Δ_h annotation. The ranking target in all three benchmarks is an orphan hypothesis, and "orphanness" is declared by the benchmark author setting `structural_additions` ≥ 1 on the candidate and specifying disconnected latents. Once those annotations are set, Proposition 1 (the mechanical consequence we proved) guarantees the ranking outcome whenever $\lambda > \kappa^*/|\Delta_h|$. Section 6 is therefore *not* an empirical discovery that D detects orphans; it is a quantitative check that the numerical condition (4) holds on the constructed benchmarks. A true test of D as

a *descriptive* tool would require hypotheses whose orphanness is *discovered* by the metric rather than declared; we do not provide such a test here. Future work: wire an LLM-driven NL-to-PyMC pipeline [Wong et al., 2023] into the engine's front end so that Δ_h is extracted from the hypothesis's natural-language statement rather than annotated by hand.

Differentiation from classical BIC is partial, not universal. The baseline comparison (§5) shows D disagreeing with full-BIC on 2 of 3 benchmarks. This is genuinely informative: when the orphan claim adds well-fitting auxiliary evidence, whole-model BIC rewards the fit while D's target-latent-specific drift does not. However, on Alice-Charlie the two criteria agree perfectly. The two are not always distinguishable on a given benchmark, and reporting D-rankings without the BIC comparison is insufficient.

Social-science benchmarks use soft proxies. Latents such as `ritual_role` are $[0, 1]$ proxies whose quantitative interpretation historians would legitimately want to scrutinise. The metric's verdict is robust to the specific proxy values (confirmed by $\times 2 \times 0.5$ sensitivity grids we ran but do not tabulate), but the benchmark itself is not a forensic reconstruction.

Authorial rather than multi-rater expert rankings. Our $\rho = 1.00$ agreement is with the benchmark author's expected ordering. Multi-rater evaluation on the historical-social cases — by actual historians of early-modern Japan and Eastern Han China, respectively — is an outstanding piece of work. We note that the system has no access to the expert ranking at inference time, so the agreement is not tautological.

Qualitative-claim ranking within "in-vocabulary". In the Noh and Eastern Han benchmarks the ordering of in-vocabulary claims partly reflects their *prior-mean distance* from the target, rather than purely their coherence contribution. This is arguably a feature (consistency with strong existing evidence = low KL) but may surprise casual readers.

Multi-latent generalisation not exercised beyond the conjugate case. The formulation of D extends immediately to a vector-valued target $\theta^* \in \mathbb{R}^d$ via the full-covariance Gaussian-moment KL estimator, but benchmarks leveraging this (e.g. the 3-parameter Dugong growth curve and the 4-parameter Peregrine-falcon population model from the boxing-gym suite) have not been exercised in this paper. Our implementation supports them out of the box.

10 Conclusion

The contribution of this paper is narrow: a PPL-native, target-latent-specific decomposition of model-selection score into two additive pieces, $D = D_{\mathrm{KL}}(\theta^* \mid h) + \lambda|\Delta_h|$, implemented against PyMC with a reference `ClaimRankingEngine`, and the honest observation that on benchmarks whose orphan adds well-fitting auxiliary evidence, D and full-BIC diverge in ranking precisely because of this target-specificity. D is *not* a new information-theoretic criterion; it is within the Friston / MDL / BIC family and its quantitative content is at most a Δ-BIC under a target-scope restriction. What we offer is the explicit decomposition, the condition under which it is guaranteed to rank orphans last (Proposition 1), and a concrete PPL-native implementation.

The evaluation is small and constructed: three hand-built benchmarks with 4 hypotheses each, authorial expected orderings, and one genuinely discriminating baseline (D vs BIC on Noh and Eastern Han). The paper's principal limitation is that we do not exhibit a case where the metric *discovers* a previously-unflagged orphan in a dataset the author did not construct. That is the next step; until then, the current state of the work is best read as a precise specification of a tool, accompanied by one non-trivial empirical differentiation from the standard baseline.

References

Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. On the logic of theory change: Partial meet contraction and revision functions. *Journal of Symbolic Logic*, 50(2):510–530, 1985.

Anonymous. MEIS: Reference implementation of the minimum embedding distance metric and benchmark suite, 2026. Code and test suite released with this paper.

Eli Bingham, Jonathan P. Chen, Martin Jankowiak, et al. Pyro: Deep universal probabilistic programming. *Journal of Machine Learning Research*, 20:1–6, 2019.

Karl Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

Xinbo Gao, Bing Xiao, Dacheng Tao, and Xuelong Li. A survey of graph edit distance. *Pattern Analysis and Applications*, 13(1):113–129, 2010.

Noah D. Goodman and Andreas Stuhlmüller. The design and implementation of probabilistic programming languages, 2014. dippl.org.

Andrew D. Gordon, Thomas A. Henzinger, Aditya V. Nori, and Sriram K. Rajamani. Probabilistic programming. *Future of Software Engineering Proceedings*, pages 167–181, 2014.

Peter D. Grünwald. *The Minimum Description Length Principle*. MIT Press, 2007.

Matthew D. Hoffman and Andrew Gelman. The no-u-turn sampler: Adaptively setting path lengths in hamiltonian monte carlo. *Journal of Machine Learning Research*, 15(1):1593–1623, 2014.

Kun-Peng Huang, Maria Ren, Chuang Li, et al. Popper: Llm-driven hypothesis falsification via sequential testing. *arXiv preprint arXiv:2502.09858*, 2025.

Hirofumi Katsuno and Alberto O. Mendelzon. Propositional knowledge base revision and minimal change. *Artificial Intelligence*, 52(3):263–294, 1991.

Imre Lakatos. Falsification and the methodology of scientific research programmes. In Imre Lakatos and Alan Musgrave, editors, *Criticism and the Growth of Knowledge*, pages 91–196. Cambridge University Press, 1970.

Judea Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, 1988.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2000.

Paolo Perrone. Categorical information geometry. *arXiv preprint*, 2024.

Du Phan, Neeraj Pradhan, and Martin Jankowiak. Composable effects for flexible and accelerated probabilistic programming in numpyro. *arXiv preprint arXiv:1912.11554*, 2019.

Willard Van Orman Quine. Two dogmas of empiricism. *The Philosophical Review*, 60(1):20–43, 1951.

Jorma Rissanen. Modeling by shortest data description. *Automatica*, 14:465–471, 1978.

John Salvatier, Thomas V. Wiecki, and Christopher Fonnesbeck. Probabilistic programming in python using pymc3. *PeerJ Computer Science*, 2:e55, 2016.

Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978.

Paul M. Williams. Bayesian conditionalisation and the principle of minimum information. *British Journal for the Philosophy of Science*, 31(2):131–144, 1980.

Lionel Wong, Gabriel Grand, Alexander K. Lew, Noah D. Goodman, Vikash K. Mansinghka, Jacob Andreas, and Joshua B. Tenenbaum. From word models to world models: Translating from natural language to the probabilistic language of thought. *arXiv preprint arXiv:2306.12672*, 2023.